



University of Science and Technology Oran Mohamed-Boudiaf
Faculty of Mathematics and Computer Science



COD-AI-CRM

Design and Development of an Intelligent CRM integrating
Machine Learning for E-Commerce

Presented By :

RezaiguiaSoltane Tadj Eddine (DS)

Khelifi Ayyoub (CS)

BellatrecheMohamed Amine (DS)

BrahimSoheib (AI)

Supervisor:

Dr. Khellat-Kihel Souad

Host:

BAT PROJET ENGINEERING

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Presentation Outline

01

Introduction & Business Context

- Host Organization
- Algerian COD Problem
- Project Objectives

02

System Architecture

- CRM Structure
- Microservices Design
- Database Overview

03

AI & Machine Learning Integration

- Risk Prediction
- Customer Segmentation
- Demand Forecasting

04

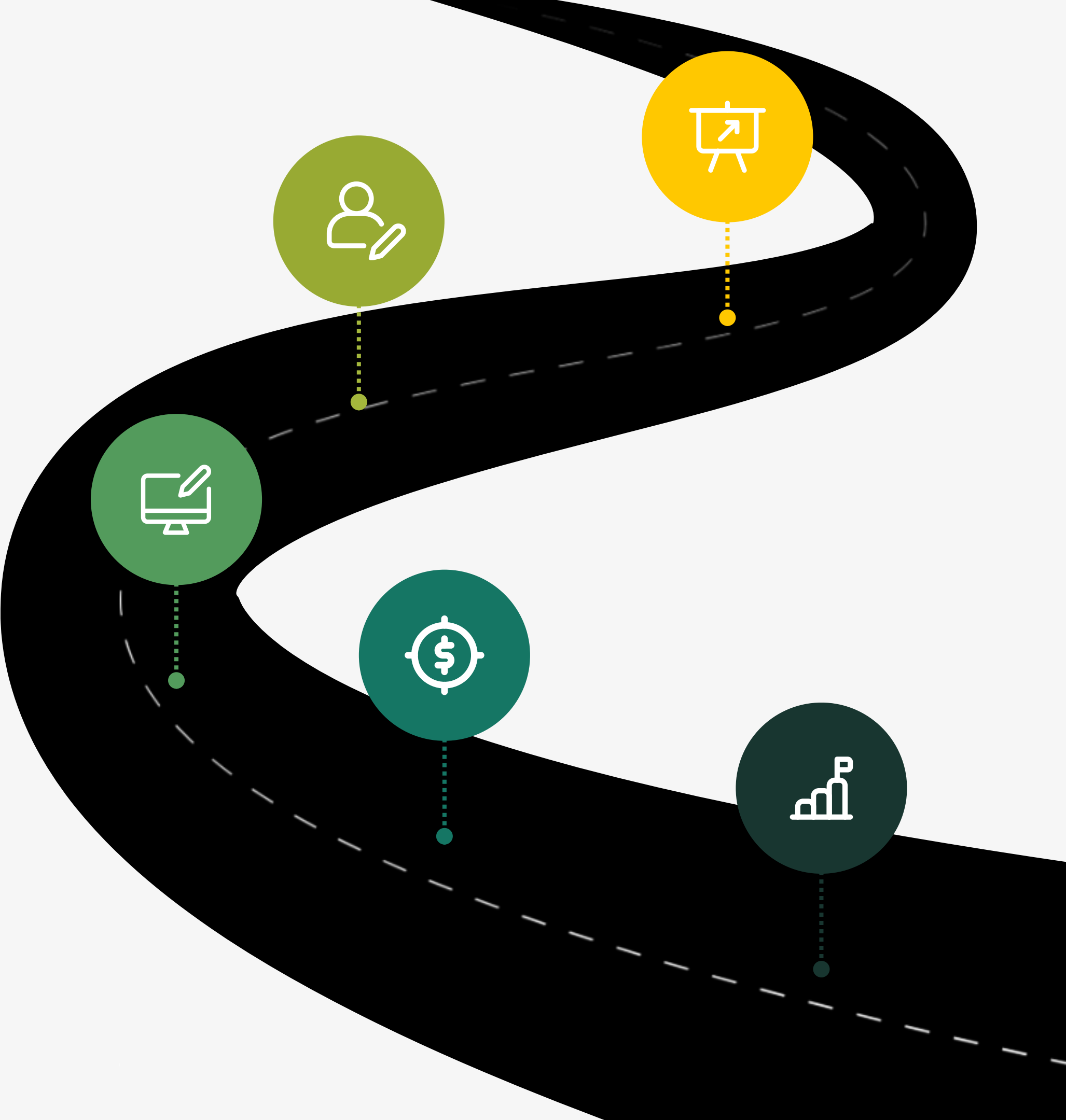
Security & Protection Layers

- Security Audit
- Vulnerabilities & Fixes
- Multi-tenant Protection

05

Demonstration & Results

- CRM Workflow
- Dashboard Demo
- Performance Metrics



Nos Solutions ERP

Une gamme complète adaptée à chaque secteur d'activité



Stock



Production



Vente



Achat



Trésorerie



Qualité



Pointage



CRM



Contrôle



Projet



Paie & RH



Portail



Import



Logistique



Comptabilité



Amortissement



Maintenance



Archive

THE ALGERIAN COD E-COMMERCE PROBLEM

04

90%+ COD Transactions:

Vast majority of Algerian e-commerce relies on Cash-on-Delivery payments.

30–50% Delivery Failure Rate:

High rates of customer unreachability and order refusals cause significant losses.

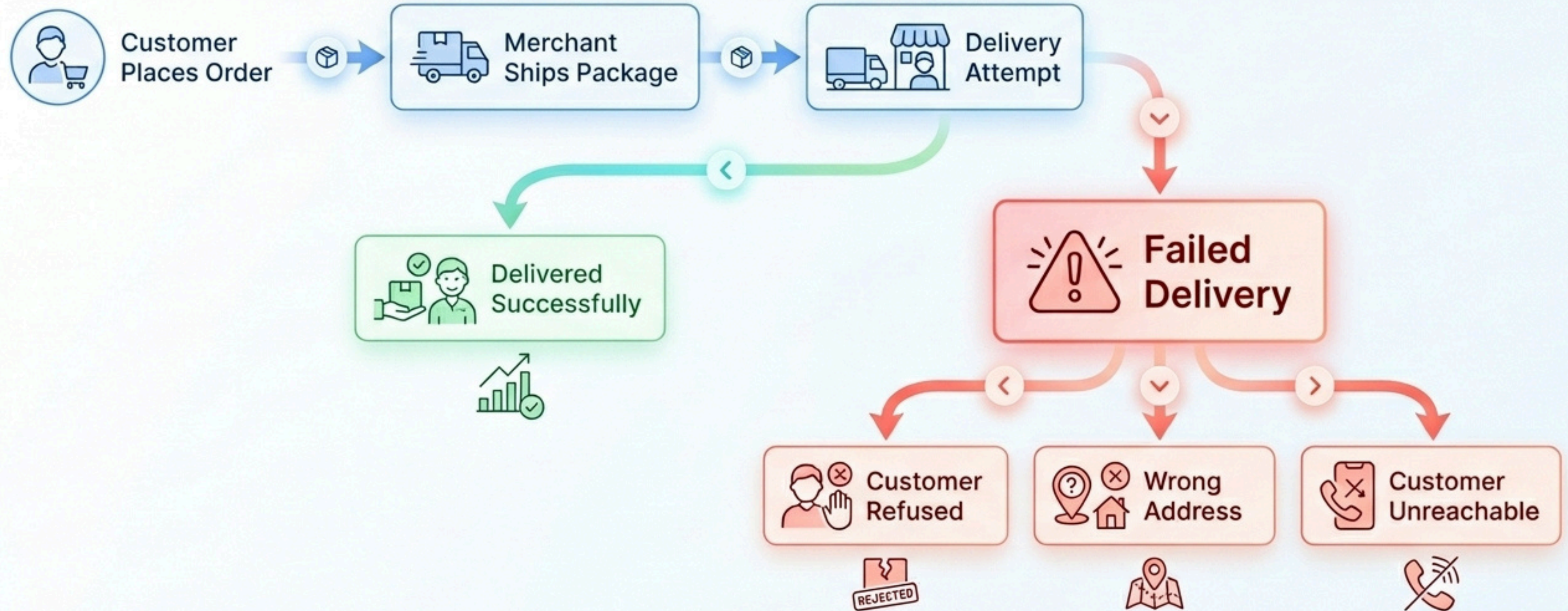
Manual Verification Inefficiencies:

Time-consuming manual processes for customer verification slow operations.

Logistics Complexity:

Coordination challenges across 69 wilayas without predictive tools.

COD Delivery Failure Workflow



Why Existing CRMs Fail vs Our Solution

Built for the Algerian COD Market

TRADITIONAL CRM LIMITATIONS



1 Generic Workflows

Designed for international online payment systems.



2 No COD Intelligence

No delivery risk prediction or customer verification logic.



3 Weak Local Adaptation

No support for wilaya logistics, Ramadan, or Eid seasonality.



4 Manual Operations

Heavy reliance on phone calls and human verification.



Traditional CRM systems are not adapted to the Algerian COD ecosystem.

FROM
Manual & Reactive Systems



TO
Intelligent & Predictive Decision-Making



COD-AI-CRM

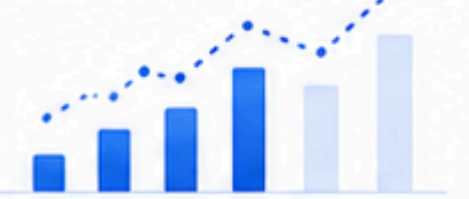
1 Risk Prediction

Predict failed deliveries before shipping.



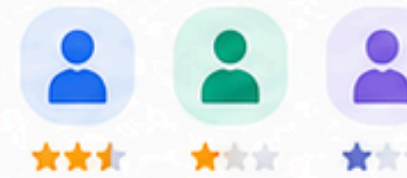
2 Demand Forecasting

Anticipate future sales and seasonal demand.



3 Customer Segmentation

Automatically classify customer profiles.



4 Cybersecurity Layers

Protect the infrastructure using multiple security mechanisms.



- ✓ Encryption
- ✓ Access Control
- ✓ Threat Detection

COD-AI-CRM
Intelligent CRM Platform



An intelligent CRM platform designed specifically for the Algerian COD market.



Built for COD Reality
Addressing real challenges of Algerian businesses.



AI-Powered Intelligence
Predict, forecast and make smarter decisions.



Higher Performance
Reduce failed deliveries and increase revenue.



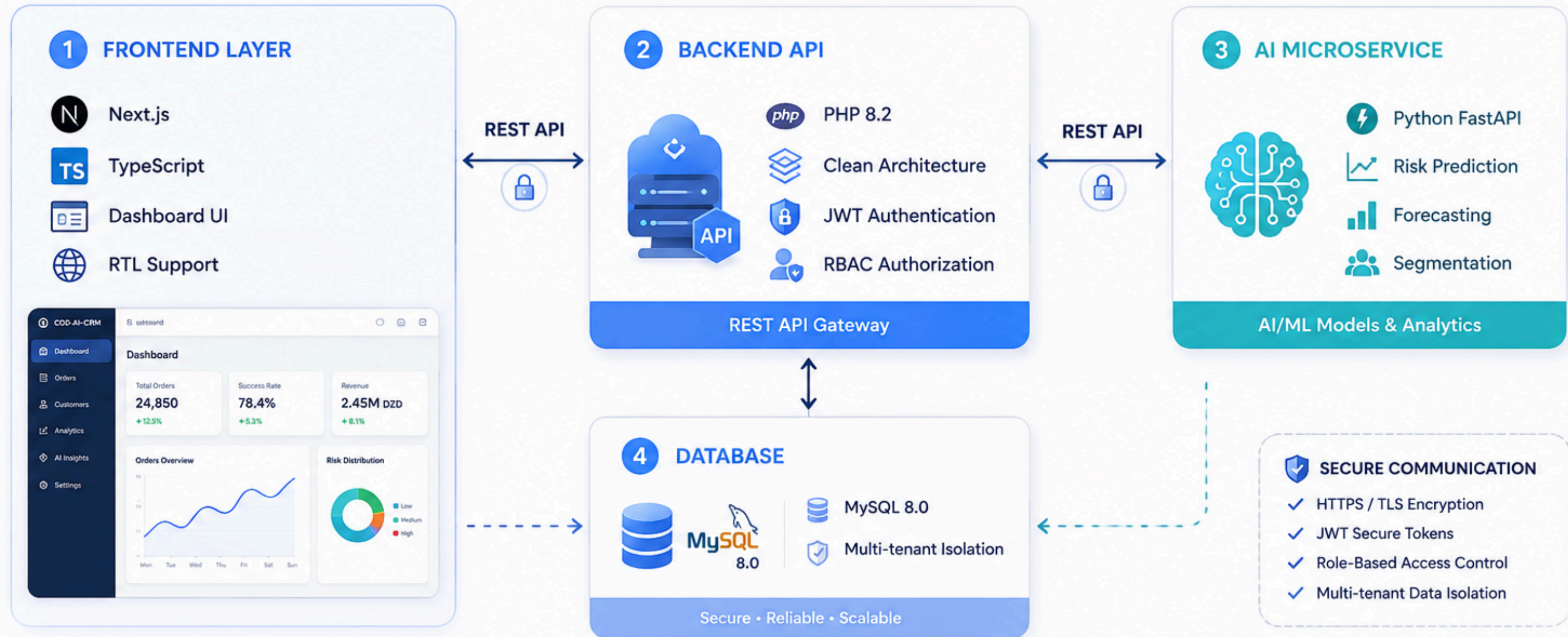
Secure by Design
Multi-layer security for safe and reliable operations.



Scalable & Multi-tenant
Designed to grow with multiple businesses.

COD-AI-CRM Microservices Architecture

Intelligent • Secure • Scalable • Multi-tenant



CLOUD READY
Deploy anywhere,
scale everywhere



MICROSERVICES
Independent services,
easy to scale



HIGH AVAILABILITY
Fault tolerant
& resilient



PERFORMANCE
Optimized APIs
& caching



OBSERVABILITY
Logs, metrics
& monitoring



Scalable and modular enterprise architecture.



AI Decision Workflow in COD-AI-CRM



CRM FEATURES AND FUNCTIONALITIES

09

Key capabilities of the COD-AI-CRM intelligent platform:



01

Predictive Delivery Risk Alerts

02

Demand Forecasting

03

Communication Automation

04

Automated Customer Segmentation

05

LLM Insights (Gemini)

TRAINING DATA: OLIST BRAZILIAN E-COMMERCE

Core constraint — no native Algerian COD dataset available

99,441

Orders Prepared

9

CSV Files Merged

1:27

BRL → DZD Rate

58

Wilayas Mapped

Geography

27 Brazilian states → 58 Algerian wilayas (zone_1, zone_2, zone_3)

Currency

Brazilian Real (BRL) → Algerian Dinar (DZD) at fixed rate 1:27

Statuses

Olist statuses → 9 COD statuses: new, confirmed, shipped, delivered...

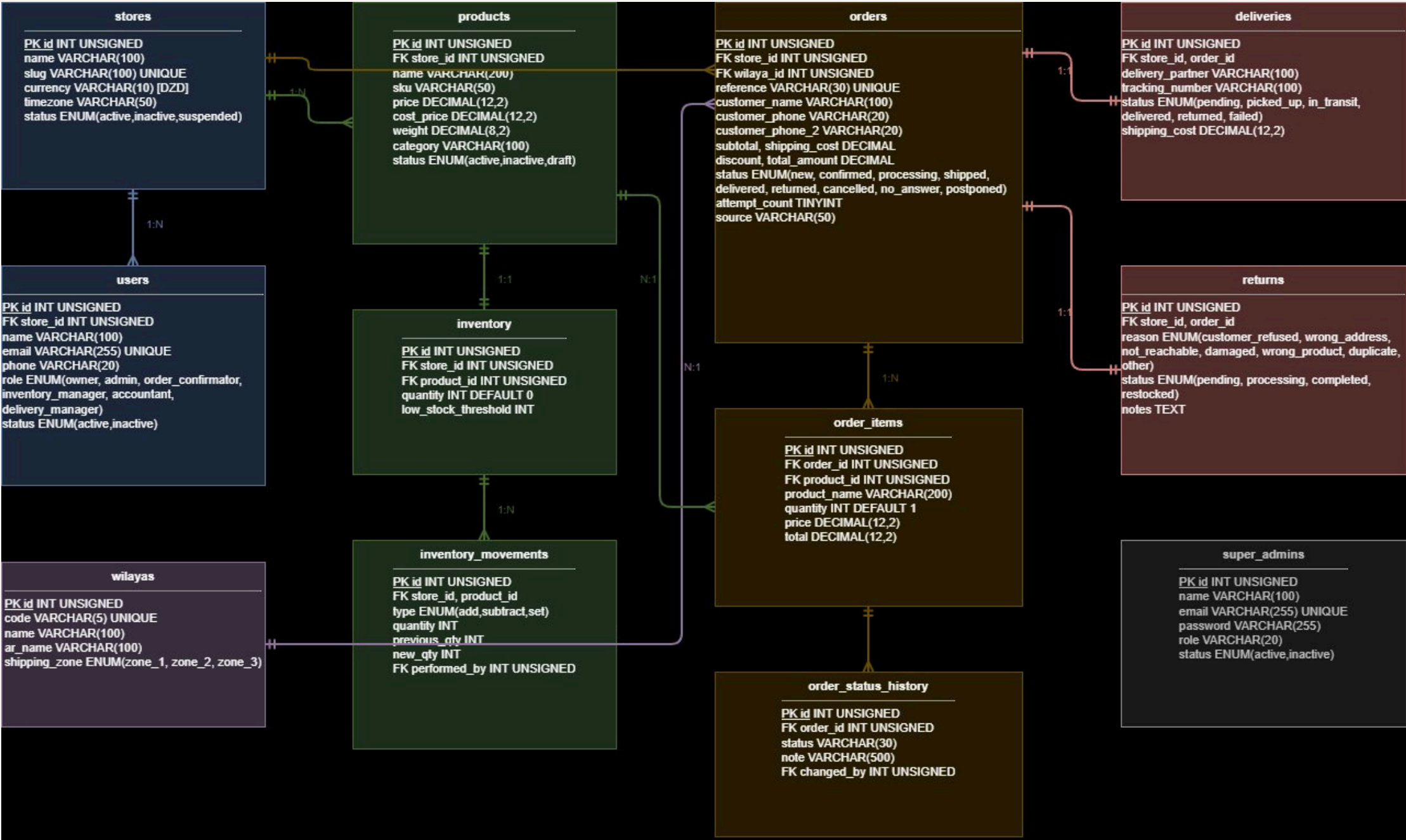
Categories

Portuguese labels translated to English + label cleaning applied

◆ The system is designed for one-click retraining on the company's real Algerian data.

DATABASE MODELING

◆ 12 tables · MySQL 8.0 · Multi-tenant architecture



orders

9 ENUM statuses incl. no_answer & postponed — COD-specific

wilayas

58 wilayas · 3 pricing zones · Arabic & Latin names

order_status_history

Full audit trail: agent, date, note at every transition

users (RBAC)

6 roles: owner, admin, confirmateur, stock, comptable, livraisons

AI INTEGRATION: FOUR FLOWS

12

◆ Prediction · Forecasting · Segmentation · LLM Insights

Flow 1 — Risk Prediction

New order → customer history enrichment → CatBoost + LightGBM + XGBoost → risk score + category

Flow 2 — Demand Forecasting

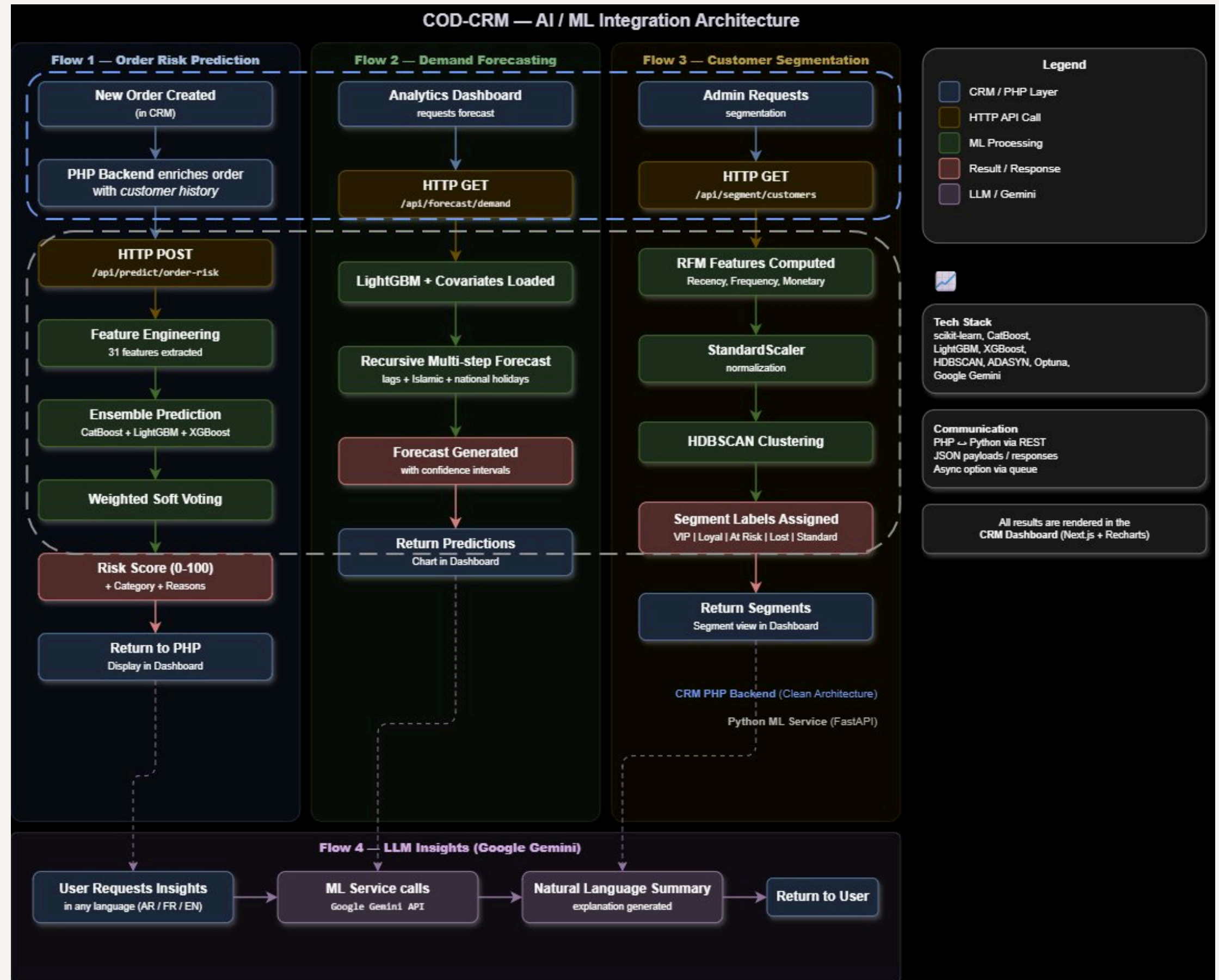
Dashboard → LightGBM + Islamic calendar covariates → recursive multi-step forecast with confidence intervals

Flow 3 — Customer Segmentation

RFM → StandardScaler → HDBSCAN / KMeans → VIP / Loyal / At-Risk / Lost / Standard labels

Flow 4 — LLM Insights (Gemini)

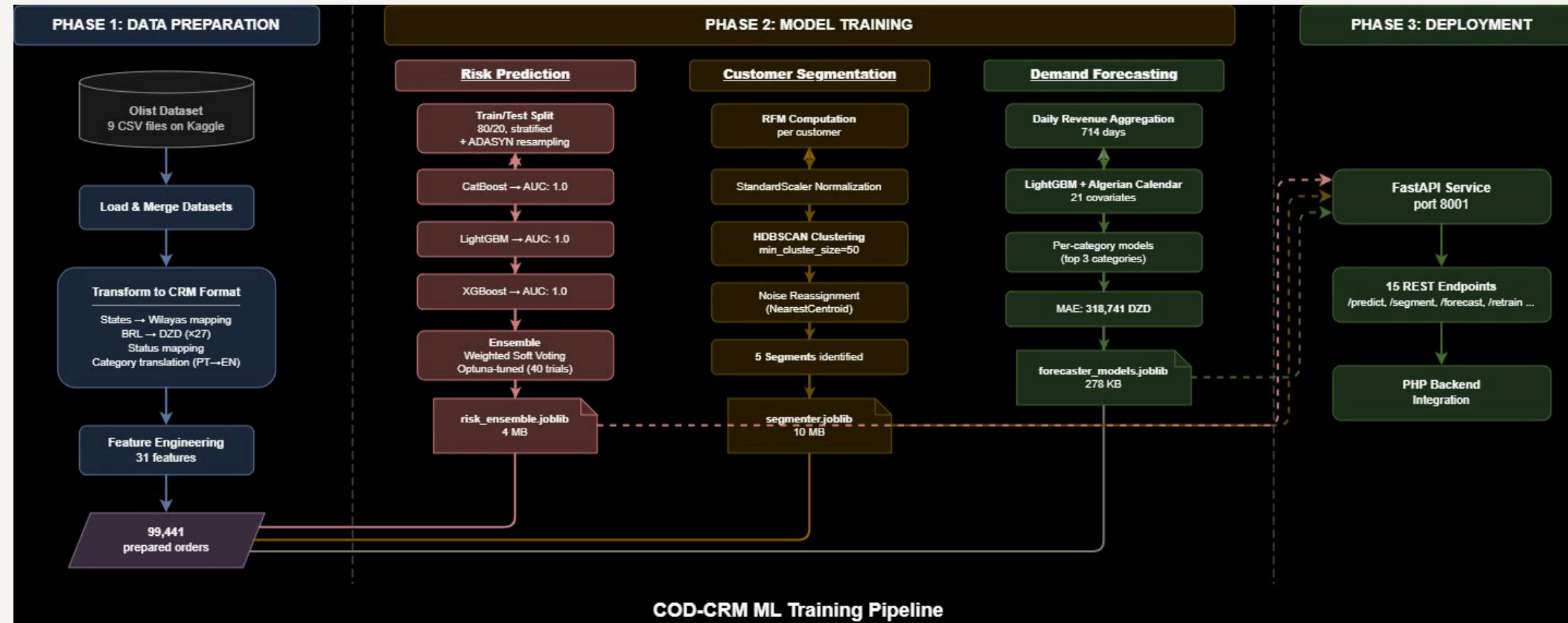
Metrics data → Google Gemini 2.5 Flash → natural language summary in AR / FR / EN



AI INTEGRATION: FOUR FLOWS

13

◆ 3 phases: Preparation → Training → Deployment



1

Phase 1 • Preparation

9 Olist CSVs merged · wilaya mapping · 31 features engineered · 99,441 orders ready

2

Phase 2 • Training

Risk ensemble 4 MB · Segmenter 10 MB · Forecaster 278 KB — all versioned

3

Phase 3 • Deployment

FastAPI on port 8001 · 15 REST endpoints · PHP backend integration

RISK PREDICTION: FEATURES & DATA LEAKAGE

14

◆ 31 features across 8 categories · Binary classification: delivered vs. failed

Temporal (6)

Hour, day, month, Algerian public holidays

Value (4)

mount, delivery cost, value/cost ratio

Customer (3)

Recurrence, average basket, order count

Payment (6)

COD, card, debit, voucher, installments

Product Quality (5)

photos count, description length, weight, volume

Geography (3)

Regional volume, seller-to-customer proximity

Category (2)

average category price, popularity index

Delivery (1)

Estimated delivery delay in days

Data Leakage detected and corrected: 3 features derived from customer/region/category success rates were artificially inflating AUC-ROC to 1.0. They were removed and the model was cleanly retrained.

OPTIMIZATION JOURNEY — AUC-ROC: 0.695 → 0.9961

15

Baseline — 0.695

20 features · SMOTE · raw target label

Clean Target — 0.827

Final statuses only (in-progress excluded) · +13.2%

31 Features · No Leakage — 0.983

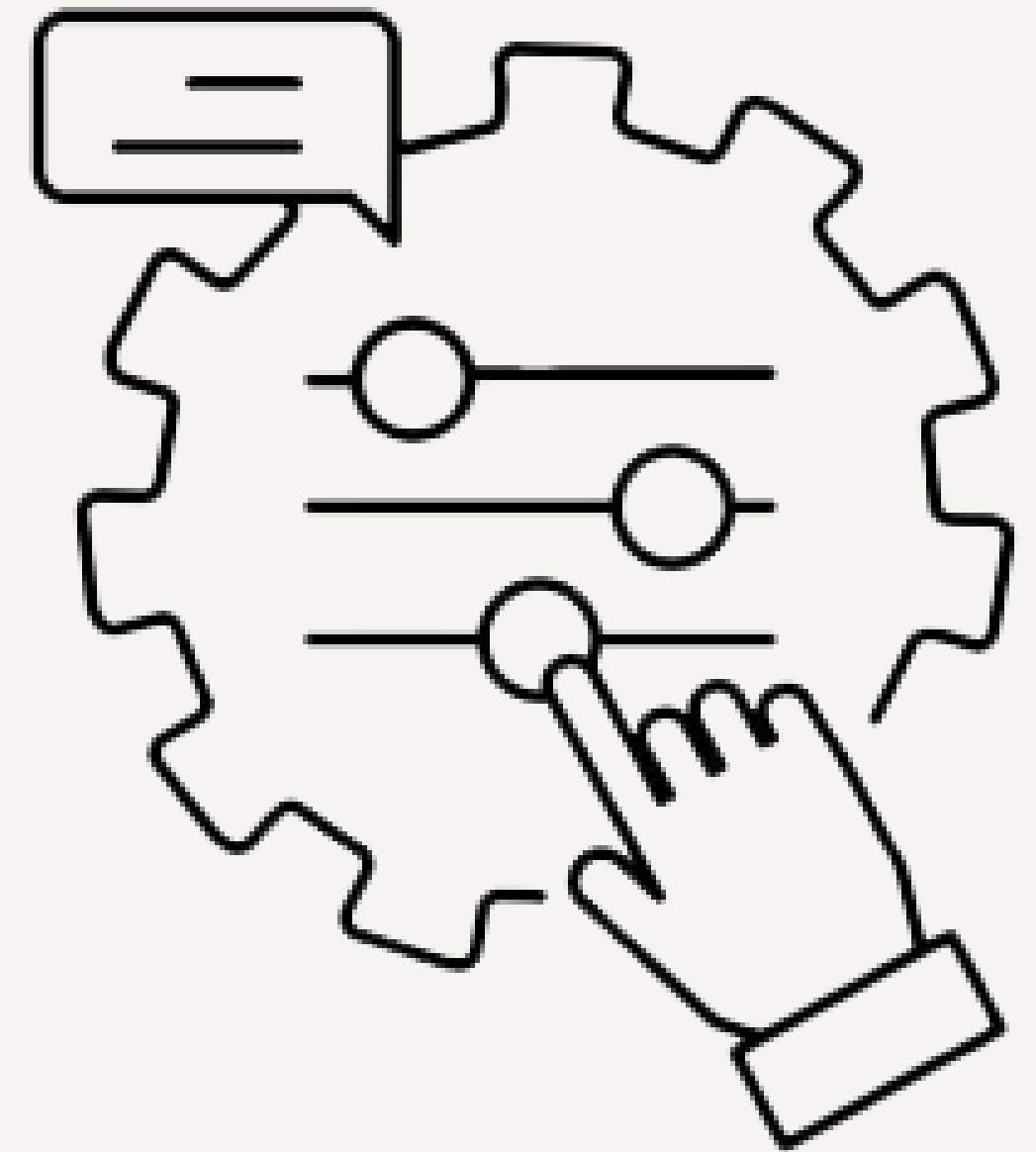
Leakage removed, enriched features added · +19.0%

ADASYN (ratio=0.3) — 0.994

Train-only adaptive oversampling vs. SMOTE · +18.9%

Optuna 80 Trials — 0.9961

Bayesian search · n_est=1289, lr=0.018, depth=12, leaves=99 · +1.1%



MODEL COMPARISON RESULTS

◆ Test set: 19,543 orders · Stratified 80/20 split · ADASYN + Optuna

Model	AUC-ROC	Accuracy	Precision	Recall	F1-Score
CatBoost	0.9957	0.9997	0.9997	1.0000	0.9999
LightGBM	0.9974	0.9997	0.9997	1.0000	0.9999
XGBoost	0.9967	0.9996	0.9997	0.9999	0.9998
Ensemble ★	0.9961	0.9997	0.9997	1.0000	0.9999

242/247

Failures Detected

Out of 247 real failures — Recall = 98%

5

False Alerts Only

Precision = 99.97%

0

Missed Failures (FN)

Zero failed orders slipped through

Decision threshold: Youden's J statistic ($J = TPR + TNR - 1$) → optimal threshold = 0.9805 — adapted to COD cost asymmetry.

DEMAND FORECASTING – LIGHTGBM + ALGERIAN CALENDAR

◆ Benchmarked against 5 approaches · 30-day horizon · Out-of-sample evaluation

Best MAE

318,741 DZD — **+12.2%** improvement vs. baseline

Why Not Foundation Models?

TimesFM / Chronos / MOIRAI are zero-shot with no exogenous covariates — ill-suited for culturally-driven demand

21 Features

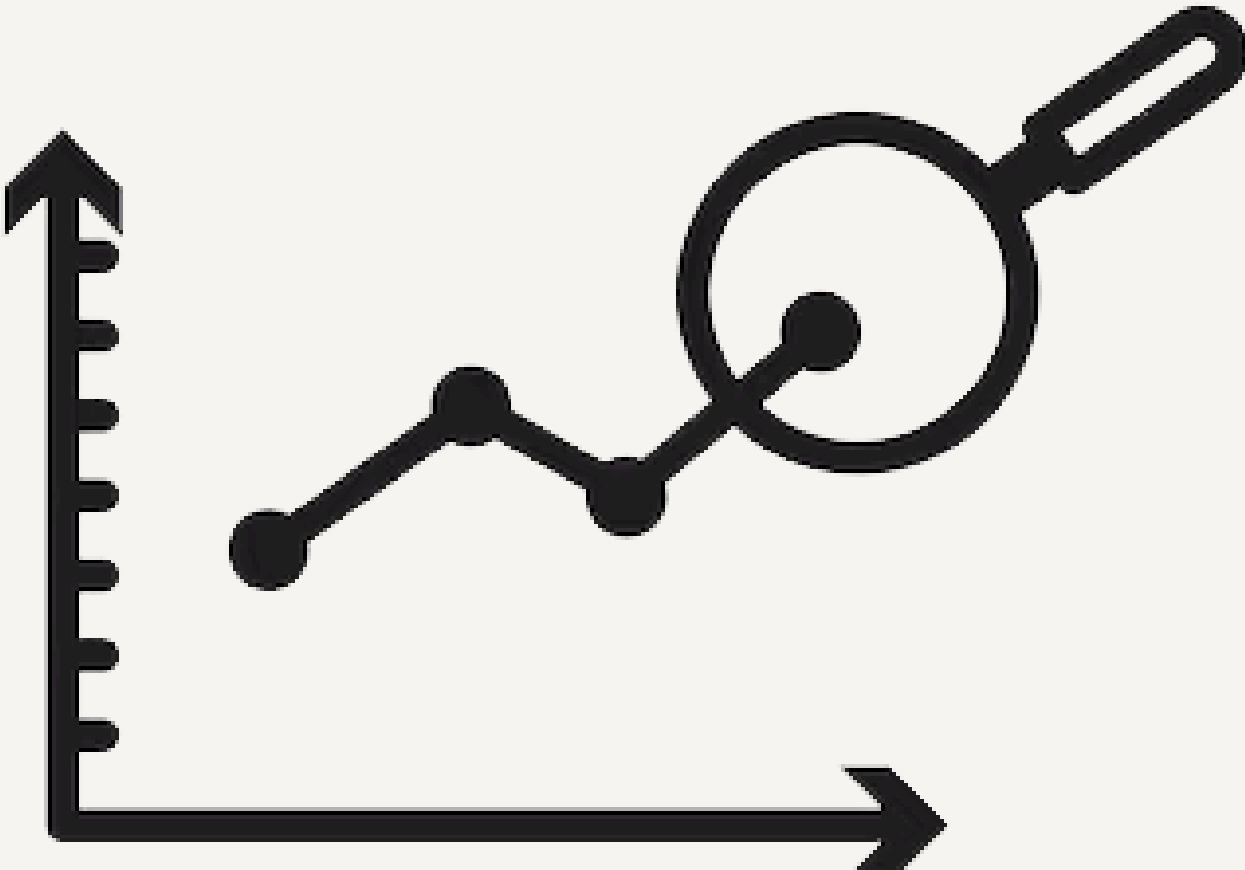
Lags (1, 7, 14, 28) · rolling stats · Ramadan · Aïd el-Fitr · Aïd el-Adha · Mawlid · national holidays

Recursive Forecasting

Each prediction is re-injected as a lag feature for the next step — 714 days of daily revenue used for training

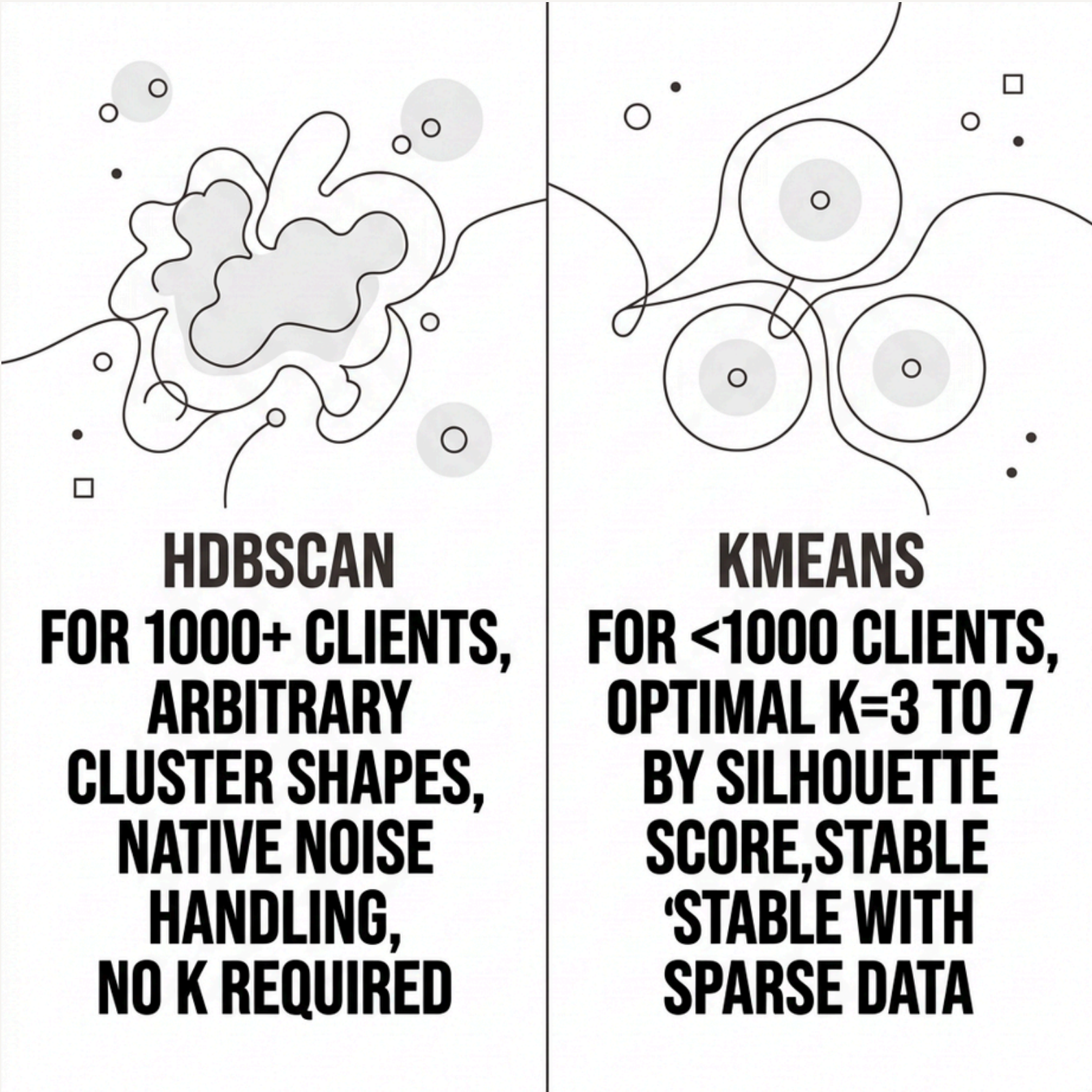
MODEL COMPARISON RESULTS

MODEL	MAE (DZD)	RMSE (DZD)	VS MA7 BASELINE
Moving Average (7d)	363,075	466,570	baseline
Prophet + Islamic holidays	376,292	447,736	-3.6%
Chronos-T5-Small (zero-shot)	338,231	395,363	+6.8%
LightGBM + lags + holidays	318,741	416,501	+12.2%



CUSTOMER SEGMENTATION – HYBRID APPROACH

◆ HDBSCAN ($\geq 1,000$ clients) · KMeans ($< 1,000$ clients) · RFM Features



Output Segments

VIP (3.8%)

Premium loyalty — priority service and exclusive offers

Loyal

Consistent buyers — nurture with engagement campaignss

At-Risk (2.8%)

Declining activity — targeted reactivation campaigns

Lost (0.5%)

9 ENUM statuses incl. no_answer & postponed — COD-specific

Standard

General base — broad conversion-focused actions

SECURITY & CYBERSECURITY ASPECTS

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Comprehensive security measures protecting the COD-AI-CRM platform:



- 01 Data Encryption
- 02 Authentication & RBAC
- 03 Threat Protection
- 04 Privacy Compliance
- 05 Continuous Monitoring

OFFENSIVE SECURITY AUDIT

4
Critical

2
Meduim

3
High

9
Total

20

Tools used: Static review (PHP code · TypeScript · Python) · Burp Suite (HTTP interception · IDOR · CSRF) · sqlmap (automated SQL injection) · IDOR tests (cross-tenant IDs manipulation)

ID	Sévérité	Problème	Correction
V-01	critical	Gemini API key committed in clear on Git	Espace réservé Suppression + .env.example
V-02	critical	ML Endpoints without authentication	Middleware Key API (X-API-Key)
V-03	critical	SQL injection in pagination (sqlmap)	Bound LIMIT/OFFSET via parameterized PDO
V-04	critical	IDOR: access to other tenants' data	Scoping store id
V-05	High	Lack of CSRF protection	CsrfMiddleware + intercepteur Axios
V-06	High	No throughput limit (brute-force)	Rate limiting 5 req/min auth · 60/min API
V-07	critical	Upload files without validation	Extension · MIME · taille 50 Mo · nom
V-08	High	Dashboard rendered without a valid session	AuthGuard Component on all routes
V-09	Meduim	Debug actif prod · Crash React unmanaged	APP_DEBUG=false · ChunkErrorRecovery

LAYERED SECURITY ARCHITECTURE

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7 levels of protection after corrections



Frontend — AuthGuard
Verification + JWT refresh before any dashboard rendering



SQL — PDO paramétré systématique
No variable interpolation—linked LIMIT/OFFSET included



Erreurs React — Error boundary + recovery
ChunkErrorRecovery—automatic reload on JS chunk error



Backend PHP — JWT + RBAC + Tenant scoping
store_id extracted from the token, role verified by route, never from the body



Uploads — Triple validation
CSV extension · MIME type · max size 50 MB · sanitization name



Microservice ML — Clé API
X-API-Key ou Bearer—optionnel dev, mandatory production



Anti-brute force — Rate limiting par paliers
5 req/min endpoints auth · 60 req/min general API routes

IMPLEMENTATION AND DEPLOYMENT

22

The following is the implementation process that must be carried out:



01

Design Phase

02

Development Milestones

03

Testing Cycles

04

Pilot Launch

05

Next Steps

DESIGN PHASE

◆ Architecture decisions · Technology choices · Schema design · API contracts

System Architecture

- Microservices: PHP backend · Python ML · Next.js frontend
- REST/JSON communication between all three components
- Docker containerization for the ML microservice
- MySQL 8.0 with multi-tenant isolation (store_id)

Technology Selection

- PHP 8.2 — Clean Architecture (Domain / App / Infra / Presentation)
- Next.js 14 — ShadCN UI, Recharts, TanStack Query
- FastAPI + Uvicorn — 15 REST endpoints
- CatBoost · LightGBM · XGBoost · HDBSCAN

Database & API Design

- 12-table schema — COD-specific statuses (no_answer, postponed)
- Audit trail on every order status change
- JWT authentication + RBAC (6 roles)
- API contracts defined before development started

◆ Design decisions were documented via draw.io schemas (architecture, DB schema, ML pipeline, AI integration flows) before writing a single line of code.

DEVELOPMENT MILESTONES

◆ Built in phases — each delivering a working increment

Phase 1 — Core CRM	Phase 2 — ML Microservice	Phase 3 — AI Integration	Phase 4 — Security & Retrain
· 11 backend modules · Multi-tenant BDD (12 tables) · JWT + RBAC authentication · Orders / Products / Inventory	· FastAPI service (port 8001) · Risk prediction ensemble · HDBSCAN segmentation · LightGBM forecasting	· PHP ↔ Python REST calls · Risk score in order workflow · Dashboard analytics + charts · Google Gemini LLM insights	· Offensive security audit · 9 vulnerabilities fixed · One-click retrain feature · Optuna hyperparameter tuning

◆ Each phase was independently testable — the ML microservice runs standalone and the CRM works without AI features if the Python service is down.

TESTING CYCLES

25

- ◆ Three parallel tracks of testing ran throughout the project

ML Model Testing

Train / Test split

80/20 stratified — 97,712 orders for training, 19,543 for testing

Data leakage audit

3 features found to inflate AUC-ROC to 1.0 artificially — removed and retrained

Benchmark comparison

5 forecasting models compared on 3 horizons (14, 30, 60 days) out-of-sample

Class imbalance handling

SMOTE tested first, then ADASYN (ratio=0.3, train-only) — ADASYN selected

Hyperparameter optimization

Optuna 80 Bayesian trials on LightGBM — n_est=1289, lr=0.018, depth=12

Security Testing (Offensive Audit)

- Static code review — PHP · TypeScript · Python
- Burp Suite — HTTP interception, IDOR, CSRF tests
- sqlmap — automated SQL injection on pagination params
- Manual IDOR — cross-tenant resource access attempts

→ **9 vulnerabilities found & fixed (4 critical · 3 high · 2 medium)**

Functional & Integration Testing

- End-to-end order workflow (new → confirmed → shipped → delivered)
- ML retrain pipeline: DB extraction → CSV → FastAPI → metrics.json
- Multi-tenant isolation: store A cannot access store B resources
- API response validation on all 15 ML endpoints

PILOT LAUNCH

◆ Platform deployed and tested with the adapted Algerian data

99,441

Orders in trained dataset

0.9961

Final AUC-ROC achieved

9

Vulnerabilities fixed

What was validated during the pilot:

Full order lifecycle

Orders created, confirmed, shipped, delivered and returned — all status transitions logged with full audit trail

Risk score in real workflow

Each new order triggers the ML microservice — score returned in <200ms, action recommendation displayed to agent

One-click retrain

Database extraction → CSV generation → Optuna tuning → new model loaded — entire cycle without developer intervention

Multilingual interface

Dashboard fully functional in Arabic (RTL), French and English — switched per user preference

◆ The retrain feature is designed so the company can adapt all models to their real Algerian COD data with a single click — no developer needed.

CONCLUSION & NEXT STEPS

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Roadmap

- Retraining on real Algerian COD data
- Integration of commercial events (back-to-school, promotions)
- Online learning for continuous model adaptation
- A/B testing of ML impact on delivery success rate

We will now walk you through a **live demonstration** of the application — showcasing each module, AI flow, and dashboard in action.

What Was Built

- **CRM Platform:** 12 tables · 9 COD statuses · 11 modules · AR/FR/EN + RTL interface
- **ML:** AUC-ROC 0.9961 · Recall 98% · FN = 0 · Segmentation 3–5 profiles · Forecast +12.2%
- **Security:** 9 vulnerabilities patched (4 critical) · 7-layer defense in depth
- **Maintainability:** One-click retraining + Optuna auto-tuning · rollback · metrics.json tracking

QUESTIONS?

Contact:

COD-AI-CRM Team

**Supervisor:**

Dr. Khellat-Kihel Souad
